

Using Age of Information for Throughput Optimal Spectrum Sharing

Hongjae Nam, Vishrant Tripathi, and David J. Love

Abstract—We consider a spectrum sharing problem where two users attempt to communicate over N channels. The Primary User (PU) has prioritized transmissions and its occupancy on each channel over time can be modeled as a Markov chain. The Secondary User (SU) needs to determine which channels are free at each time-slot and attempt opportunistic transmissions. The goal of the SU is to maximize its own throughput, while simultaneously minimizing collisions with the PU, and satisfying spectrum access constraints. To solve this problem, we first decouple the multiple-channel problem into N single-channel problems. For each decoupled problem, we prove that there exists an optimal threshold policy that depends on the last observed PU occupancy and the freshness of this occupancy information. Second, we establish the indexability of the decoupled problems by analyzing the structure of the optimal threshold policy. Using this structure, we derive a Whittle index-based scheduling policy that allocates SU transmissions using the Age of Information (AoI) of accessed channels. We also extend our insights to PU occupancy models that are correlated across channels and incorporate learning of unknown Markov transition matrices into our policies. Finally, we provide detailed numerical simulations that demonstrate the performance gains of our approach.

I. INTRODUCTION

Spectrum sharing plays a fundamental role in modern wireless communication systems, particularly in the context of 6G networks, which aim to achieve ultra-reliable and low-latency communications (URLLC) while efficiently utilizing the radio spectrum. Effective spectrum sharing allows secondary users (SUs) - unlicensed or lower-priority users - to access available spectrum, enabling *opportunistic spectrum* utilization without interfering with primary users (PUs) - licensed or higher-priority users. This dynamic access approach maximizes spectral efficiency while supporting adaptive and intelligent resource allocation strategies [1]–[3].

Fig. 1 describes a typical spectrum sharing scenario. The Primary User (PU), such as a naval radar and communications system, transmits prioritized information. A Secondary User (SU) tries to utilize unused channels to send lower-priority information. The Citizens Broadband Radio Service (CBRS), in the 3.55-3.7 GHz frequency range, serves as a representative example of such spectrum sharing. The band consists of three kinds of users - incumbent users (such as naval radars or satellite systems), Priority Access License (PAL) users (typically 4G LTE and 5G operators), and General

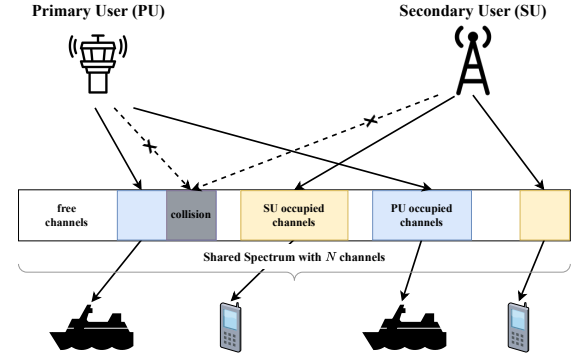


Fig. 1. A spectrum sharing scenario with a primary user (PU) and a secondary user (SU) attempting transmissions over N shared channels. The SU needs to carefully decide which channels to utilize so as to avoid collisions with the PU while still maximizing its own throughput.

Authorized Access (GAA) users. Incumbent and PAL users (PUs) are protected from interference from GAA users (SUs) through restrictions on simultaneous channel usage. In this tiered access system, the PU has the flexibility to select and utilize its preferred channels, and its spectrum usage pattern may vary dynamically over time. Consequently, the SU must continuously identify open channels for opportunistic spectrum access while minimizing interference with the PU's transmissions.

In this paper, we address how the SU should go about making spectrum access decisions in a dynamic setting while optimizing its own throughput and minimizing collisions with the PU, in the presence of spectrum access constraints. We start by modeling the PU's channel occupancies as independent Markov chains. In this simplified setting, we analyze structural properties of optimal spectrum access policies and design a low-complexity near-optimal scheme that utilizes both historical PU occupancy information and its freshness to decide which channels to utilize next. We then extend our results to more general channel occupancy models that are correlated across channels, and incorporate learning of unknown Markov transition matrices into our policies.

Optimal spectrum access policies, as we will show later, will depend on the last known channel occupancy information and their Age of Information (AoI). This is because of our Markov modeling assumptions. Recent observations are far more informative than older ones in making decisions about the future. AoI, a metric that quantifies the freshness of information, has gained significant attention in the context of wireless networks [4]–[9]. The interplay between accurate

H. Nam, V. Tripathi, and D. J. Love are with the Elmore Family School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN 47907 USA (e-mail: {nam86, tripathi, djlove}@purdue.edu). This work was supported in part by the Office of Naval Research (ONR) under Grant N000142112472, and the National Science Foundation (NSF) under Grants CNS2212565, CNS2225578, and EEC1941529.

monitoring of channels and throughput maximization leads to an interesting tradeoff: minimizing monitoring error (i.e., AoI) often necessitates transmissions across many channels, while optimizing throughput might mean utilizing the channels that are most likely to be immediately free.

Related Work: Spectrum sharing strategies have been extensively studied in the literature, with a focus on maximizing throughput while minimizing interference. Traditional approaches employ multi-armed bandit (MAB) models or partially observable Markov decision process (POMDP)-based formulations to model the uncertain availability of channels [10]–[13]. Notably, myopic sensing policies [10], [11] have been shown to achieve near-optimal performance in a two-channel scenario and provided numerical results suggesting their efficiency in multi-channel settings. However, their theoretical optimality has only been proven for independent two-channel cases, and the analysis focuses solely on maximizing throughput without considering collision penalties or access constraints in the presence of hierarchical users.

Queue-based models have also been employed in analyzing spectrum access [12], [13]. However, these approaches assume that the channel occupancies follow independent Bernoulli processes, which overlook temporal correlation in channel states and traffic patterns. In addition to queue focused analyses, AoI has also emerged as a key metric for evaluating the freshness of channel state information (CSI) in time-varying channel environments [14]–[16].

Closer to our work, the authors in [17]–[20] have applied the Whittle index to spectrum sharing. In these prior works, the focus is primarily on the discounted reward problem. In addition, the derived Whittle index expressions typically involve infinite series and must be approximated or truncated for computation. In this work, we derive closed-form computable expressions for the Whittle index in the average reward setting and use our insights to tackle more complicated scenarios involving correlation and learning of channel states. *Ours is the first work to explicitly highlight the close connection between AoI and optimal spectrum sharing.*

Contributions: In this work, we propose a novel approach to spectrum sharing that integrates Markov channel occupancies with AoI, allowing us to develop throughput optimal spectrum access policies.

- 1) We formulate spectrum sharing as an optimization problem that maximizes SU throughput subject to collision penalties and spectrum access constraints.
- 2) We decouple this problem into N single-channel problems through a Lagrangian relaxation and demonstrate that the optimal policy has a threshold structure.
- 3) We establish indexability of the original problem and use the optimal thresholds to develop the Whittle index as a closed-form function of AoI, enabling low-complexity near-optimal performance.
- 4) We extend our approach to handle correlated PU channel occupancies by designing an AoI-based heuristic index utilizing insights from the Whittle index analysis. We also incorporate learning of unknown Markov transition

matrices.

Organization: The rest of the paper is organized as follows. In Section II, we describe the general system model and introduce the spectrum sharing problem. In Section III, we solve the decoupled single-channel problem via Bellman equations and propose the optimal threshold-type policy. In Section IV, we derive a closed-form expression for the Whittle index using the optimal threshold policy and propose a Whittle index-based scheduling algorithm for the multi-channel setting. Section V extends our Whittle index approach to scenarios involving inter-channel correlation. In Section VI, we provide simulation results to validate our analysis.

II. SYSTEM MODEL

We consider a time-slotted network with two classes of users trying to transmit information over shared spectrum. The *Primary User* (PU) gets priority when transmitting on any channel and does not have any spectrum access constraints. The *Secondary User* (SU) needs to find channels on which the PU is not currently transmitting, so as to be able to communicate. The SU pays a cost for colliding with the PU or for occupying too many channels. The goal of the SU is to maximize its total time-average throughput, while ensuring that it does not collide with the PU, and satisfy its own energy/spectrum access constraints.

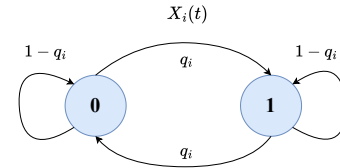


Fig. 2. A two-state symmetric Markov chain that models the PU occupancy of the i -th channel.

We assume that there are N channels available in total. We model the PU occupancy on each channel through a symmetric two-state Markov chain as shown in Fig. 2. If channel i is currently occupied by the PU, then in the next time-slot, it remains occupied with probability q_i and becomes free with probability $1 - q_i$. This allows us to model historical dependence of channel occupancy. Specifically, we assume that $q_i < 1/2, \forall i \in [N]$. Thus, if the PU is currently occupying a channel, it is more than likely to still be occupying the channel in the next time-slot. We make the following key assumptions in our model:

Assumption 1. *The SU knows the Markov transition probabilities $\{q_i \mid i \in [N]\}$ for each channel i in the spectrum and the corresponding transition matrices are given by*

$$Q_i = \begin{bmatrix} 1 - q_i & q_i \\ q_i & 1 - q_i \end{bmatrix}, \forall i \in [N]. \quad (1)$$

Assumption 2. *The PU channel occupancies, modeled by 2-state Markov chains, evolve **independently** across channels.*

These assumptions will allow us to perform analysis, provide performance guarantees, and gain important insights into

the process of spectrum sharing. Importantly, we will design spectrum access policies for the SU that utilize the knowledge of the transition matrices as well as PU channel occupancy information obtained in the past, to optimize the overall throughput of the SU. In Section V, we extend our Whittle index-based approach to scenarios where Assumption 2 is relaxed, specifically when PU channel occupancies exhibit correlation across channels. In Section VI, we also provide numerical results when Assumption 1 is relaxed, i.e., the transition matrices Q_i are unknown and need to be estimated.

Let $\mathbf{X}(t) = [X_1(t), X_2(t), \dots, X_N(t)]^T \in \{0, 1\}^N$ be the channel state vector representing PU occupancy at the time-slot t , modeled as independent two-state (Gilbert-Elliott) Markov channels [21].

Sensing and Collision Model: In each time-slot, the SU can decide whether to transmit on channel i or leave it empty. Let $u_i(t) \in \{0, 1\}$ indicate whether the SU attempts a transmission on channel i at time-slot t . Suppose the SU transmits on channel i at time-slot t ($u_i(t) = 1$). Then, if the channel is already occupied by the PU ($X_i(t) = 1$), the SU transmission leads to a collision with the PU and fails. However, if the channel is unoccupied by the PU ($X_i(t) = 0$), the SU successfully transmits one packet. The failure or success of the transmission is revealed to the SU at the beginning of the next time-slot $t + 1$ via an ACK.

Measuring Throughput: The SU successfully delivers one packet only when the PU is absent and no collision occurs. Mathematically, let $R_i(t)$ indicate whether a packet was successfully transmitted by the SU over channel i at time-slot t . Then, it is given by

$$R_i(t) = u_i(t)(1 - X_i(t)), \quad \forall i \in [N]. \quad (2)$$

Clearly, the SU transmits a packet, i.e., $R_i(t) = 1$, when it selects channel i ($u_i(t) = 1$) and the PU is absent on that channel ($X_i(t) = 0$). Otherwise, the throughput is zero.

Spectrum Access Constraints: Due to practical hardware limitations and energy constraints, the SU will typically not be capable of simultaneously sensing the entire set of available channels in the spectrum. Further, even if it were capable of doing so, attempting transmission on all channels will cause significant disruption to the PU. So, we assume that the SU can access at most L channels in each time-slot. Mathematically,

$$\sum_{i=1}^N u_i(t) \leq L, \quad \forall t \in [T], \quad (3)$$

where L is the maximum number of channels the SU can access at a given time. As a result, an efficient sensing strategy is required to intelligently select channels based on past PU occupancy observations.

Collision Penalties: The SU may unintentionally select channels that are already occupied by a PU, leading to interference with higher-priority PU transmissions. Mathematically, a collision occurs when the SU selects channel i ($u_i(t) = 1$) while the PU is also present on the same channel ($X_i(t) = 1$).

We use $C_i(t)$ to indicate the collision penalty incurred by the SU on channel i at time-slot t .

$$C_i(t) = \gamma u_i(t) X_i(t), \quad \forall i \in [N], \quad (4)$$

where $\gamma > 0$ represents the penalty cost for each individual collision. Imposing this penalty ensures that the SU's channel selection strategy considers the impact on both its own throughput and the PU's performance.

Given the setup described above, we formulate a unified optimization framework.

Problem 1 (SU Throughput Maximization)

Given Markov transition probabilities $\{q_i \mid i \in [N]\}$, a spectrum usage constraint of at most L channels at each time-slot, and a collision penalty of $\gamma > 0$ per collision, we want to find a causal policy π that solves

$$\begin{aligned} \max_{\pi} \quad & \lim_{T \rightarrow \infty} \mathbb{E}_{\pi} \left[\frac{1}{L} \frac{1}{T} \sum_{i=1}^N \sum_{t=1}^T (R_i(t) - C_i(t)) \right] \\ \text{s.t.} \quad & \sum_{i=1}^N u_i(t) \leq L, \quad \forall t \in [T], \\ & u_i(t) \in \{0, 1\}, \quad \forall i \in [N], \forall t \in [T]. \end{aligned} \quad (\text{P1})$$

Solving (P1) directly is challenging due to the spectrum access constraint in each time-slot. As our system model satisfies the Markov property, (P1) can be formulated as a restless multi-armed bandit (RMAB) problem with state transitions governed by the PU and SU. Since the RMAB problem has been proven to be *PSPACE-hard* [22], finding an optimal solution to (P1) is generally intractable. This is because our action space in each time-slot is combinatorial over the number of channels and the number of spectrum access policies scales exponentially in the length of the time-horizon, making dynamic programming (DP) based solutions infeasible. To manage this complexity, we first relax the per-time-slot constraint into a time-average constraint. We then use a Lagrangian relaxation to decouple (P1) into N single-channel problems, each of which is easier to solve optimally.

To decouple the problem across channels, we introduce a Lagrangian multiplier $C > 0$, which acts as the price for accessing a channel. Normalizing this price defines an effective transmission cost $D = (\gamma + C)/(1 + \gamma)$, which leads to the following result.

Theorem 1. *The spectrum sharing problem (P1) decouples into N single-channel problems of the form below, where $D > 0$ is an effective transmission cost.*

$$\begin{aligned} \max_{\pi} \quad & \lim_{T \rightarrow \infty} \mathbb{E}_{\pi} \left[\frac{1}{T} \sum_{t=1}^T (R_i(t) - D u_i(t)) \right] \\ \text{s.t.} \quad & u_i(t) \in \{0, 1\}, \quad \forall t \in [T]. \end{aligned} \quad (\text{P1-d})$$

We provide a proof of Theorem 1 in Appendix A of our technical report [23]. By obtaining the optimal policy for each

decoupled single-channel problem with an effective transmission cost (P1-d), we can optimally solve the overall Lagrangian relaxation. As we will discuss later, this will also allow us to design a spectrum access policy for the original problem (P1) that is nearly optimal via the Whittle index framework.

III. SOLVING THE DECOUPLED PROBLEM

In this section, we will solve the decoupled single-channel problem (P1-d) using dynamic programming. The first observation we make is that the decoupled problem can be modeled as a Markov decision process (MDP) with an infinite horizon time-average reward. To describe this single-channel MDP, we need to describe four things - the state space, the action space, the rewards, and the Markov transitions.

States: Let $\Delta_i(t)$ denote the number of time slots elapsed since the SU most recently accessed channel i as of time-slot t . Then, the SU has two pieces of information - 1) the last observed PU occupancy, i.e., $X_i(t - \Delta_i(t))$ and 2) the staleness of the PU channel occupancy information at the SU, i.e., $\Delta_i(t)$. We refer to $\Delta_i(t)$ as the AoI of channel i . Since we assume the transition probability q_i to be less than 1/2, the PU occupancy is more likely to remain unchanged than to flip after the last observation. Accordingly, the maximum a posteriori (MAP) estimate at time-slot t is simply the last observed channel occupancy: $\hat{X}_i(t) = X_i(t - \Delta_i(t))$. Thus, the state of our MDP will be the following two-dimensional tuple $\{\hat{X}_i(t), \Delta_i(t)\}$.

Actions: In each time-slot t , the SU has only one decision to make - whether to access channel i (set $u_i(t) = 1$) and pay an effective transmission cost D in the hope of throughput reward, or not transmit (set $u_i(t) = 0$).

Rewards: The reward in our MDP is straightforward. In each time-slot, the reward for channel i equals throughput minus an effective transmission cost, i.e., $R_i(t) - Du_i(t)$. More importantly, we are interested in time-average reward, rather than discounted reward.

Transitions: Given our states, actions, and rewards, Lemma 1 describes how the MDP evolves over time.

Lemma 1. *Given the state $\{\hat{X}_i(t), \Delta_i(t)\}$ at time-slot t , the actual PU occupancy at time-slot t is distributed as follows*

$$\mathbb{P}(X_i(t) = \hat{X}_i(t)) = [Q_i^{\Delta_i(t)}]_{00} \quad (5)$$

$$\mathbb{P}(X_i(t) \neq \hat{X}_i(t)) = [Q_i^{\Delta_i(t)}]_{10}. \quad (6)$$

Here $[Q_i^{\Delta}]_{mn}$ represents the (m, n) -th entry of Q_i^{Δ} , which is the Δ -th power of the matrix Q_i . Using this, we can now write down state transitions as a function of the channel access decision $u_i(t)$.

$$\Delta_i(t+1) = \begin{cases} \Delta_i(t) + 1, & \text{if } u_i(t) = 0 \\ 1, & \text{if } u_i(t) = 1. \end{cases} \quad (7)$$

$$\hat{X}_i(t+1) = \begin{cases} \hat{X}_i(t), & \text{if } u_i(t) = 0 \\ X_i(t), & \text{if } u_i(t) = 1. \end{cases} \quad (8)$$

Proof. See Appendix B in the technical report [23]. ■

As a consequence of Lemma 1, the estimated PU channel occupancy at any time can be computed as just a function of the last known occupancy, the transition matrix Q_i and the AoI. Given the MDP described above, we will now describe the Bellman equations for our decoupled single-channel problem.

Since the analytical approach is the same across all N instances of the decoupled problem (P1-d), we will omit the channel identifier i from our analysis below. Let λ denote the time-average reward under the optimal policy. Further, let $S(\hat{X}(t), \Delta(t))$ represent the differential value function, which measures the relative value of being in state $(\hat{X}(t), \Delta(t))$ compared to this time-average reward.

Lemma 2. *The Bellman Equations for the decoupled single-channel problem are given by the following two equations.*

$$S(1, \Delta) = \max_{u \in \{0,1\}} \left\{ S(1, \Delta + 1), [Q^{\Delta}]_{10}(S(0, 1) + 1) + [Q^{\Delta}]_{00}S(1, 1) - D \right\} - \lambda, \quad \forall \Delta \in \mathbb{Z}^+. \quad (9)$$

$$S(0, \Delta) = \max_{u \in \{0,1\}} \left\{ S(0, \Delta + 1), [Q^{\Delta}]_{00}(S(0, 1) + 1) + [Q^{\Delta}]_{10}S(1, 1) - D \right\} - \lambda, \quad \forall \Delta \in \mathbb{Z}^+. \quad (10)$$

Proof. See Appendix C in the technical report [23]. ■

An optimal policy must satisfy the system of equations (9) and (10) above. Before we find such a policy, we will look at policies with a threshold-type structure.

Definition 1. *A threshold policy with the threshold H for the decoupled single-channel problem computes the spectrum access decisions for the SU via the following mapping.*

$$u(t) = \begin{cases} 1, & \text{if } \hat{X}(t) = 0 \\ 1, & \text{if } \hat{X}(t) = 1 \text{ and } \Delta(t) \geq H \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

Intuitively, a threshold policy of the form above allows the SU to keep utilizing the channel when it is unoccupied by the PU ($\hat{X}(t) = 0$). As soon as it realizes that the PU is actually occupying the channel ($\hat{X}(t) = 1$), it stops transmitting and then waits for H time-slots before trying a re-transmission. The following lemma characterizes the time-average reward for such threshold policies.

Lemma 3. *The time-average reward under the threshold policy is a function of the threshold H and the effective transmission cost D .*

$$\lambda(H, D) = \frac{[Q^H]_{10} - ([Q^H]_{10} + [Q]_{10})D}{[Q^H]_{10} + H[Q]_{10}}. \quad (12)$$

Proof. See Appendix D in the technical report [23]. ■

The structure of the threshold policies is motivated by our Markov assumptions. If the SU attempted a transmission in the last time-slot and succeeded, we know that the PU had left the channel unoccupied. Since $q < 1/2$, the channel is still more likely to be unoccupied in the next time-slot, and our certainty about the channel being empty only decreases with time. So, the SU should keep transmitting until at some point in the future, when the PU comes back to this channel and causes a collision. At that point, we expect that at least for the next few time-slots, the PU will continue occupying the channel. So, the SU should wait for some threshold H before it tries re-transmission. It turns out that simply choosing the best possible threshold over this simple class of policies leads to a policy that solves (P1-d) optimally.

Theorem 2. Given $q \in (0, 1/2)$ and an effective transmission cost D , the single-channel problem is optimally solved by a threshold policy of the form (11). $H^*(q, D)$ denotes the optimal threshold and satisfies

$$g(H^*) \leq \lambda \leq g(H^* - 1) \quad (13)$$

where $g(H^*) \triangleq ([Q^{H^*+1}]_{10} - [Q^{H^*}]_{10})(S(0, 1) + 1)$. If no such H^* can be found, the optimal policy for the SU is to never transmit.

Proof. See Appendix E in the technical report [23]. ■

IV. SPECTRUM ACCESS POLICIES

In the previous section, we established that to solve the decoupled single-channel problem, an optimal policy has the threshold-type structure as described by (11). This solution can be applied to each channel i to compute an optimal threshold H_i^* , thus solving the Lagrangian relaxed problem. However, it still does not provide a solution for the original problem (P1) with strict spectrum access constraints. To this end, we introduce the Whittle index approach [24], which has been shown to be asymptotically optimal in the many-users regime for AoI scheduling problems [25] and allows us to provide a solution to the original scheduling problem.

A. Whittle Index Policy

To apply the Whittle index approach to our problem and find an optimal policy for (P1), we first need to verify a key structural property known as *indexability* for the decoupled problems.

Definition 2. Indexability [24]. Given cost D , let $\mathcal{P}(D)$ be the set of states for which the optimal action in the decoupled problem is not to transmit. The problem is indexable if the set $\mathcal{P}(D)$ monotonically increases from the empty set to the entire state space, as D increases from 0 to ∞ .

Definition 3. Whittle index [24]. Denote by $W(\hat{X}, \Delta)$ the Whittle index in state $(\hat{X}, \Delta)$. Given indexability, $W(\hat{X}, \Delta)$ is the minimum cost D that makes both actions ($u = 1$ and $u = 0$) equally desirable in state $(\hat{X}, \Delta)$.

The following lemma establishes that indexability holds for the decoupled single-channel problems.

Lemma 4. The indexability property holds for each decoupled single-channel problem (P1-d), given $q_i \in (0, 1/2)$.

Proof. See Appendix F in the technical report [23]. ■

Since the *indexability* condition holds for each decoupled problem, we can compute the Whittle index.

Theorem 3. For the i -th decoupled problem (P1-d), the Whittle index $W_i(\hat{X}_i, \Delta_i)$ is given by

$$W_i(\hat{X}_i, \Delta_i) \triangleq \begin{cases} \infty, & \text{if } \hat{X}_i = 0 \\ \frac{\Delta_i([Q_i^{\Delta_i-1}]_{10} - [Q_i^{\Delta_i}]_{10}) + [Q_i^{\Delta_i}]_{10}}{(\Delta_i - 1)([Q_i^{\Delta_i-1}]_{10} - [Q_i^{\Delta_i}]_{10}) + [Q_i^{\Delta_i}]_{10} + [Q_i]_{10}}, & \text{if } \hat{X}_i = 1. \end{cases} \quad (14)$$

Proof. See Appendix F in the technical report [23]. ■

Importantly, the Whittle index in Theorem 3 captures the semantics encoded in the SU's channel knowledge $(\hat{X}_i(t), \Delta_i(t))$, rather than the AoI $\Delta_i(t)$ alone. The Whittle index at state $(\hat{X}_i(t), \Delta_i(t))$ measures how valuable it is to schedule a transmission on that channel, given that the last observed PU occupancy was $\hat{X}_i(t)$ and this observation was made $\Delta_i(t)$ time-slots ago. Clearly, if we know that the PU is not occupying the channel ($\hat{X}_i(t) = 0$), we should attempt a transmission, which suggests the infinite Whittle index value. However, if we last observed the channel to be occupied by the PU ($\hat{X}_i(t) = 1$), the Whittle index is low for small values of AoI and gradually increases as $\Delta_i(t)$ increases, since it becomes more and more likely that the channel occupancy flipped. Using (14), we can now design the Whittle index policy in Algorithm 1 to address the original problem (P1).

Intuitively, the algorithm prioritizes channels with a higher likelihood of being free immediately and for longer periods by assigning them larger Whittle index values. This reflects the cost an SU is willing to pay or the perceived worth of channel access. The algorithm selects the top L channels based on the Whittle Index and then chooses those which exceed their optimal threshold under the collision penalty.

V. CORRELATED CHANNEL OCCUPANCIES

A drawback of our analysis and the Whittle index approach is that it does not immediately extend to settings where the PU occupancy is correlated across channels. A typical example would be the case where the PU uses a contiguous band of B channels, and the center of this band moves randomly across the available spectrum over time. To tackle such settings, we first develop a heuristic index function motivated by our intuitive understanding of the Whittle index. This heuristic index policy will then provide us with a pathway to solving the general scenario.

Algorithm 1: Whittle Index for Spectrum Sharing

Input: maximum SU channels L , collision penalty γ .**Output:** Spectrum access policy
 $u_i(t), \forall i \in [N], \forall t \in [T]$.**1 Initialization:****2** $\Delta_i(0) \leftarrow 1, \forall i \in [N]$ **3** $\hat{X}_i(0) \leftarrow 1, \forall i \in [N]$ **4 for** $t = 0$ **to** T **do****5** Sort channels by Whittle index $W_i(\hat{X}_i(t), \Delta_i(t))$ **6** For the top L channels:**7** $u_i(t) \leftarrow \begin{cases} 1, & \text{if } \Delta_i(t) \geq H_i^*(q_i, \frac{\gamma}{1+\gamma}) \\ 0, & \text{otherwise} \end{cases}$ **8** For the rest: $u_i(t) \leftarrow 0$ **9 for** $i = 1$ **to** N **do****10** $\Delta_i(t+1) \leftarrow \begin{cases} \Delta_i(t) + 1, & \text{if } u_i(t) = 0 \\ 1, & \text{if } u_i(t) = 1 \end{cases}$ **11** $\hat{X}_i(t+1) \leftarrow \begin{cases} \hat{X}_i(t), & \text{if } u_i(t) = 0 \\ X_i(t), & \text{if } u_i(t) = 1 \end{cases}$

A. Heuristic Index Policy

Let's suppose that the last known PU occupancy for channel i is 1, i.e., $\hat{X}_i(t) = 1$. Further, let's suppose that this observation is $\Delta_i(t)$ time-slots old. The following lemma characterizes how many packets the SU would be able to transmit if it chooses channel i in the next time-slot and continues transmitting until it first observes a collision.

Lemma 5. *Given channel i occupancy information $\hat{X}_i(t) = 1$ and the AoI $\Delta_i(t)$ at time-slot t , the number of packets $G_i(t)$ that the SU can transmit on channel i starting at time-slot t until it faces its first collision satisfies the following*

$$\mathbb{E}[G_i(t)] = \frac{[Q_i^{\Delta_i}]_{10}}{[Q_i]_{10}}. \quad (15)$$

Proof. See Appendix G in the technical report [23]. ■

We can think of this expectation as the potential reward for choosing channel i at time-slot t . This motivates a heuristic index policy for our spectrum access problem - where we use the following index function $V_i(\hat{X}_i, \Delta_i)$ instead of the Whittle index functions $W_i(\hat{X}_i, \Delta_i)$ derived in Theorem 3.

$$V_i(\hat{X}_i, \Delta_i) \triangleq \begin{cases} \infty, & \text{if } \hat{X}_i = 0 \\ \frac{[Q_i^{\Delta_i}]_{10}}{[Q_i]_{10}}, & \text{if } \hat{X}_i = 1. \end{cases} \quad (16)$$

By substituting (16) in place of (14) in Algorithm 1, we can implement a heuristic index policy, which leverages the simplified index for efficient spectrum access decisions. We will show later via simulations in Section VI that this heuristic index performs almost as well as the Whittle index for independent PU occupancies across the channels.

The benefits of this simpler index are evident when the PU occupancies are correlated across channels. The new index can

be reformulated as a function of the observed PU occupancy history in correlated scenarios (since observations of channel j will affect estimates for channel i even if $i \neq j$).

$V_i(\text{PU occupancy history})$

$$\triangleq \begin{cases} \infty, & \text{if } \hat{X}_i = 0 \\ \mathbb{E}[G_i(t) | \text{PU occupancy history}], & \text{if } \hat{X}_i = 1. \end{cases} \quad (17)$$

This conditional expectation is often easy to compute *or estimate using data*, and we can utilize the resulting index functions V_i to implement an index policy for correlated or data-driven scenarios. In Section VI we will show that we can effectively do this for PU occupancies that move in a contiguous block over time.

VI. NUMERICAL RESULTS

In this section, we present simulation results that validate our analytical findings on the performance of the Whittle index. The simulations are conducted with a time horizon of $T = 30000$, and each data point represents an average over 100 simulation runs. Unless specified, the Markov channels have transition probabilities evenly spaced from $q_{min} = 0.1$ to $q_{max} = 0.5$ and the collision penalty $\gamma = 0.5$.

A. Independent Channel Occupancies

Under the independent channel setting, we compare the performance of five different policies: 1) The Pure Random policy, which selects L channels uniformly at random regardless of previous selections, serves as a lower bound on performance and is similar to the random access characteristics of slotted ALOHA [26]. 2) The Check Empty + Random policy continues transmissions on channels that were found to be empty in the previous time-slot and selects a new channel uniformly at random for each scheduling decision that leads to a collision, similar to CSMA-style protocols [27]. 3) The Whittle index policy, described in Section IV, which selects channels with the highest Whittle indices when collision occurs. 4) The Heuristic index policy described in Section V, which utilizes (16) to approximate the Whittle index. 5) The Liu et al. policy, where we implement the average-reward Whittle index in [17] (i.e., the discount factor $\beta \rightarrow 1$ limit of their discounted formulation). We define the normalized throughput as the average successful SU throughput per time-slot, normalized by the number of selected channels L .

Fig. 3 depicts the impact of the selected number of channels L on system performance. From Fig. 3(a), the normalized SU throughput decreases as L increases due to higher competition, with the index policies performing best and the Pure Random policy worst. From Fig. 3(b), the average collision rate rises with L , but the Whittle index policy reduces collisions effectively. Notably, when selecting between 1 channel and $N/4$ channels, our Whittle index policy achieves up to a **19% normalized SU throughput gain** and a **39% collision rate reduction** compared to the Check Empty + Random policy. The Heuristic index policy also closely approximates the performance of the Whittle index. Notably, the Liu et al.

average-reward Whittle index [17], is nearly identical to our Whittle index because their first crossing time corresponds to our AoI state under perfect sensing.

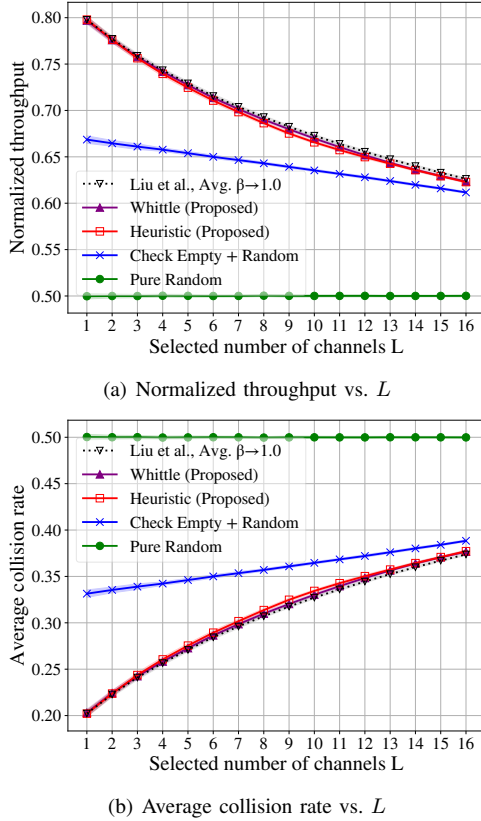


Fig. 3. Performances of different scheduling policies vs. selected channel number L under independent channel occupancies (fixed total channel number $N = 32$).

Finally, we emphasize that our Whittle index is written explicitly in closed form as a function of AoI, unlike prior works [17]–[20]. By avoiding implicit quantities, this explicit AoI form provides a clear intuition for the heuristic index and practical extensions to unknown Markov transitions and correlated-channel scenarios in the subsequent sections.

B. Estimating Unknown Markov Transition Probabilities

We now repeat the numerical simulation from Fig. 3 but with two variations of our proposed policies, that do not know the underlying Markov transition matrices Q_i in advance. So along with the four algorithms already plotted, we add the Whittle index policy with learning, and the Heuristic index policy with learning. For these, the transition probabilities q_i are not known a priori and instead estimated over time using transmission outcomes.

The online estimate of the transition probabilities $\hat{q}_i(t)$ can be computed as a function of the number of transitions of the channel state ($0 \rightarrow 0$ and $0 \rightarrow 1$). Specifically

$$\hat{q}_i(t) = \frac{N_i^{0 \rightarrow 1}(t)}{N_i^{0 \rightarrow 1}(t) + N_i^{0 \rightarrow 0}(t)}, \quad (18)$$

where $N_i^{\hat{X} \rightarrow \hat{Y}}(t)$ gives the number of observed transitions of channel i 's occupancy from state $\hat{X}$ to state $\hat{Y}$ up to time t .

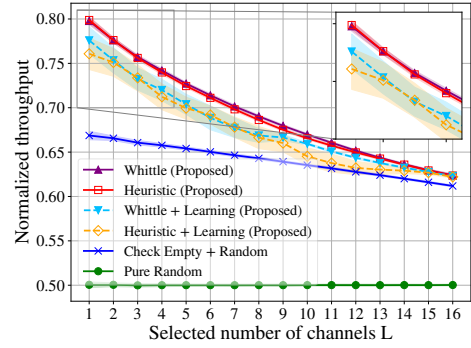


Fig. 4. Performances of different scheduling policies with and without online estimation of $\hat{q}_i$.

Fig. 4 shows that the learning-based Whittle and heuristic policies maintain performance levels that closely approximate their non-learning counterparts. Notably, the Whittle + Learning policy achieves up to a **16% normalized SU throughput gain** compared to the Check Empty + Random policy. The Heuristic + Learning policy also closely approximates the performance of the Whittle + Learning policy due to the online estimation of q_i values.

C. Correlated Channel Occupancies

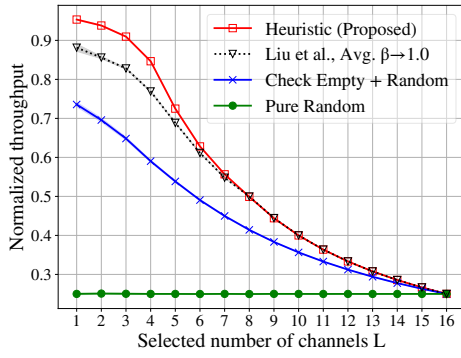
We now study the correlated channel setting. Here, we compare the performance of four different policies: 1) The Pure Random policy, 2) The Check Empty + Random policy, 3) The Heuristic index policy described in Section V, which utilizes (17) for computing heuristic indices from PU occupancy history with cross-channel correlation, and 4) The Liu et al. policy, where we estimate a per-channel effective independent transition probability from observed state flips and plug it into the independent Markov model assumed in [17]. We assume that the PU occupies a contiguous band of $B = 12$ channels out of a total of $N = 16$ available channels, and the center position of this band follows a discretized Gaussian random walk over time, with noise variance $\sigma^2 = 1$.

To find an expression for the correlated index based on (17), we approximate the expected free duration of channel i given prior observation history using mean first-passage times for Brownian motion. The full derivation of the correlated index is provided in Sec. VI of our technical report [23].

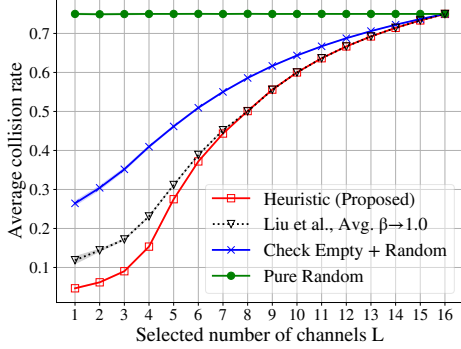
Fig. 5 depicts the performance of different policies as L varies. In particular, compared to the Check Empty + Random policy, our heuristic index policy achieves up to a **43% gain in normalized SU throughput** and an **82% reduction in the collision rate** when the number of selected channels ranges from 1 to the maximum 4 available, significantly outperforming it in the most critical operational range, where scheduling is necessary. When compared to the Liu et al. policy, the heuristic index attains up to a 10% throughput gain and a 61% collision rate reduction, highlighting the benefits of our AoI and channel semantics based approach.

VII. CONCLUSION

In this work, we studied spectrum sharing where a secondary user opportunistically communicates over multiple



(a) Normalized throughput vs. L



(b) Average collision rate vs. L

Fig. 5. Performances of different scheduling policies vs. selected channel number L under correlated channel occupancies.

Markov channels occupied by a primary user. We established indexability and derived the Whittle index policy with an explicit AoI functional form, which demonstrates strong performance by achieving both high SU throughput and low collision rates. Furthermore, we extended our approach to accommodate PU occupancy models with inter-channel correlation and learning of unknown Markov matrices. Important directions of future work involve 1) proving theoretical guarantees for the heuristic index policy in scenarios with correlated channels, 2) considering multiple SUs and competition between them for open channels, and 3) extending data-driven online learning and prediction for more general PU occupancy models.

REFERENCES

- [1] J. Mitola, "Cognitive radio for flexible mobile multimedia communications," in *1999 IEEE International Workshop on Mobile Multimedia Communications (MoMuC'99)(Cat. No. 99EX384)*. IEEE, 1999, pp. 3–10.
- [2] R. Ramanathan and C. Partridge, "Next generation (XG) architecture and protocol development (XAP)," *DARPA, New York, Tech Rep. AFRL-IF-RS-TR-2005-281*, 2005.
- [3] C. G. Brinton, M. Chiang, K. T. Kim, D. J. Love, M. Beesley, M. Repeta, J. Roesse, P. Beming, E. Ekudden, C. Li *et al.*, "Key focus areas and enabling technologies for 6G," *IEEE Communications Magazine*, vol. 63, no. 3, pp. 84–91, 2025.
- [4] S. Leng and A. Yener, "Impact of imperfect spectrum sensing on age of information in energy harvesting cognitive radios," in *ICC 2019 - 2019 IEEE International Conference on Communications (ICC)*, 2019, pp. 1–6.
- [5] —, "Minimizing age of information for an energy harvesting cognitive radio," in *2019 IEEE Wireless Communications and Networking Conference (WCNC)*, 2019, pp. 1–6.
- [6] V. Tripathi and E. Modiano, "A whittle index approach to minimizing functions of age of information," *IEEE/ACM Transactions on Networking*, pp. 1–15, 2024.
- [7] V. Tripathi, L. Ballotta, L. Carlone, and E. Modiano, "Computation and communication co-design for real-time monitoring and control in multi-agent systems," in *2021 19th International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt)*. IEEE, 2021, pp. 1–8.
- [8] I. Kadota, E. Uysal-Biyikoglu, R. Singh, and E. Modiano, "Minimizing the age of information in broadcast wireless networks," in *2016 54th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE, 2016, pp. 844–851.
- [9] I. Kadota, A. Sinha, E. Uysal-Biyikoglu, R. Singh, and E. Modiano, "Scheduling policies for minimizing age of information in broadcast wireless networks," *IEEE/ACM Transactions on Networking*, vol. 26, no. 6, pp. 2637–2650, 2018.
- [10] Q. Zhao and B. Krishnamachari, "Structure and optimality of myopic sensing for opportunistic spectrum access," in *2007 IEEE International Conference on Communications*. IEEE, 2007, pp. 6476–6481.
- [11] Q. Zhao, B. Krishnamachari, and K. Liu, "On myopic sensing for multi-channel opportunistic access: structure, optimality, and performance," *IEEE Transactions on Wireless Communications*, vol. 7, no. 12, pp. 5431–5440, 2008.
- [12] T. Stahlbuhk, B. Shrader, and E. Modiano, "Throughput maximization in uncooperative spectrum sharing networks," in *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2016, pp. 1242–1246.
- [13] —, "Throughput maximization in uncooperative spectrum sharing networks," *IEEE/ACM Transactions on Networking*, vol. 28, no. 6, pp. 2517–2530, 2020.
- [14] S. Farazi, A. G. Klein, and D. R. Brown, *The Age of Channel State Information*. Cambridge University Press, 2023.
- [15] —, "Bounds on the age of information for global channel state dissemination in fully-connected networks," in *2017 26th International Conference on Computer Communication and Networks (ICCCN)*. IEEE, 2017, pp. 1–7.
- [16] A. G. Klein, S. Farazi, W. He, and D. R. Brown, "Staleness bounds and efficient protocols for dissemination of global channel state information," *IEEE Transactions on Wireless Communications*, vol. 16, no. 9, pp. 5732–5746, 2017.
- [17] K. Liu and Q. Zhao, "Indexability of restless bandit problems and optimality of Whittle index for dynamic multichannel access," *IEEE Transactions on Information Theory*, vol. 56, no. 11, pp. 5547–5567, 2010.
- [18] K. Liu, Y. Zhang, and Z. Ding, "Efficient algorithm design of dynamic spectrum access by Whittle index," *arXiv preprint arXiv:2501.00236*, 2024.
- [19] J. Nino-Mora, "A restless bandit marginal productivity index for opportunistic spectrum access with sensing errors," in *International Conference on Network Control and Optimization*. Springer, 2009, pp. 60–74.
- [20] R. Meshram, D. Manjunath, and A. Gopalan, "On the whittle index for restless multiarmed hidden markov bandits," *IEEE Transactions on Automatic Control*, vol. 63, no. 9, pp. 3046–3053, 2018.
- [21] E. N. Gilbert, "Capacity of a burst-noise channel," *Bell system technical journal*, vol. 39, no. 5, pp. 1253–1265, 1960.
- [22] C. H. Papadimitriou and J. N. Tsitsiklis, "The complexity of optimal queuing network control," *Mathematics of Operations Research*, vol. 24, no. 2, pp. 293–305, 1999.
- [23] H. Nam, V. Tripathi, and D. J. Love, "Using age of information for throughput optimal spectrum sharing," *arXiv preprint arXiv:2509.18465*, 2025.
- [24] P. Whittle, "Restless bandits: Activity allocation in a changing world," *Journal of applied probability*, vol. 25, no. A, pp. 287–298, 1988.
- [25] A. Maatouk, S. Kriouile, M. Assad, and A. Ephremides, "On the optimality of the whittle's index policy for minimizing the age of information," *IEEE Transactions on Wireless Communications*, vol. 20, no. 2, pp. 1263–1277, 2021.
- [26] N. Abramson, "The aloha system: Another alternative for computer communications," in *Proceedings of the November 17-19, 1970, fall joint computer conference*, 1970, pp. 281–285.
- [27] L. Kleinrock and F. Tobagi, "Packet switching in radio channels: Part I-carrier sense multiple-access modes and their throughput-delay characteristics," *IEEE transactions on Communications*, vol. 23, no. 12, pp. 1400–1416, 1975.